

Fraud Signal Report

Statistical Analysis of UPI Fraud Detection Dataset

Objective

The objective of this project is to analyze UPI transaction data using statistical techniques to identify fraud patterns, detect suspicious transaction behavior, and provide business recommendations that can help fraud prevention teams improve transaction security and reduce fraudulent activities.

Dataset Overview

- **Dataset Name:** UPI Fraud Detection Dataset
- **Total Records:** 26,394
- **Total Features:** 65
- **Target Variable:** is_fraud

The dataset contains transaction details, customer information, device characteristics, authentication behavior, account information, and several fraud-related indicators used for statistical analysis.

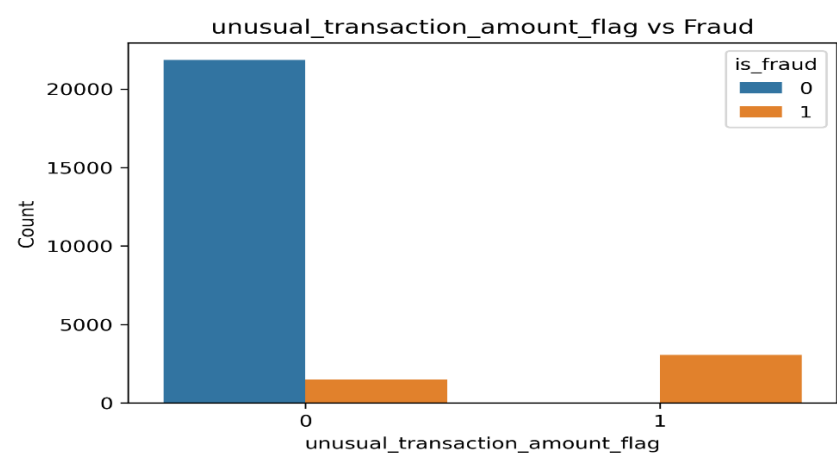
Key Fraud Signals

The statistical analysis identified several important fraud indicators that contribute significantly to fraudulent transactions.

- High Transaction Amount
 - Unusual Device Usage
 - Unusual IP Address
 - Unusual Location
 - Rapid Transaction Velocity
 - Multiple Failed Authentication Attempts
 - Frequent OTP Requests
 - Newly Created Receiver Accounts
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Fraud Distribution

Chart 1: Fraud Distribution

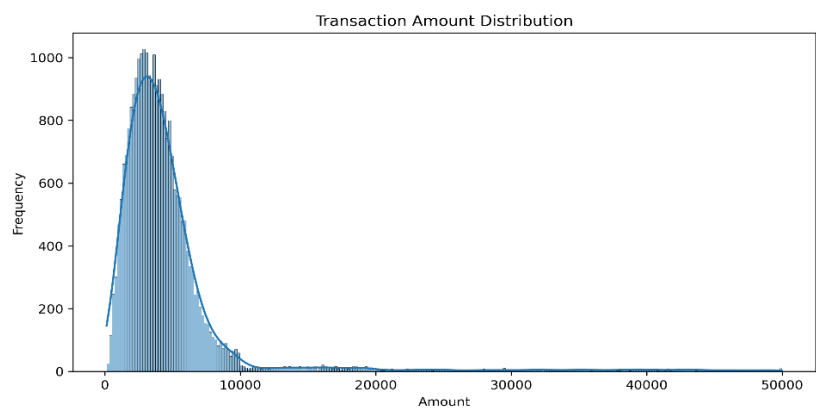


Interpretation

The fraud distribution chart shows the proportion of fraudulent and legitimate transactions in the dataset. Legitimate transactions represent the majority of the dataset, while fraudulent transactions form a smaller portion. This imbalance is common in real-world fraud detection datasets and highlights the importance of accurately identifying fraudulent cases without generating excessive false alarms.

Transaction Amount by Fraud Status

Chart 2:

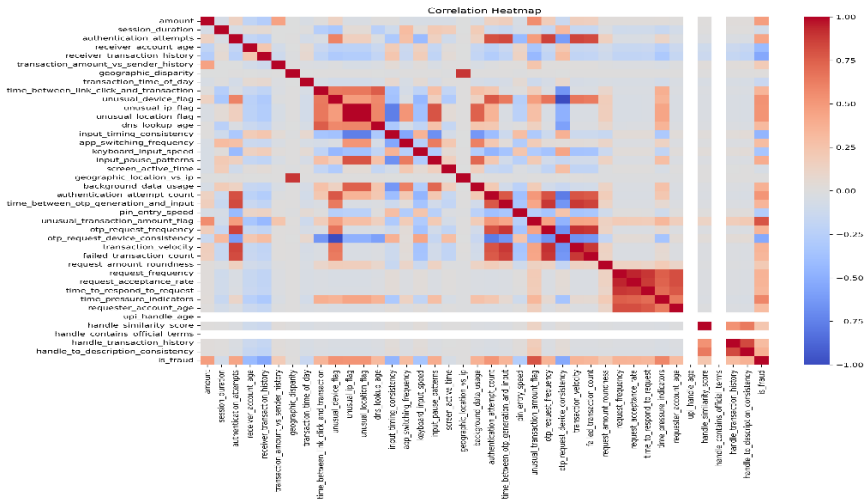


Interpretation

The boxplot indicates that fraudulent transactions generally involve higher transaction amounts than legitimate transactions. However, high-value transactions alone are not sufficient to classify fraud, as many genuine customers also perform high-value transactions. Transaction amount should therefore be combined with additional fraud indicators during risk assessment.

Correlation Analysis

Chart 3: Correlation Heatmap

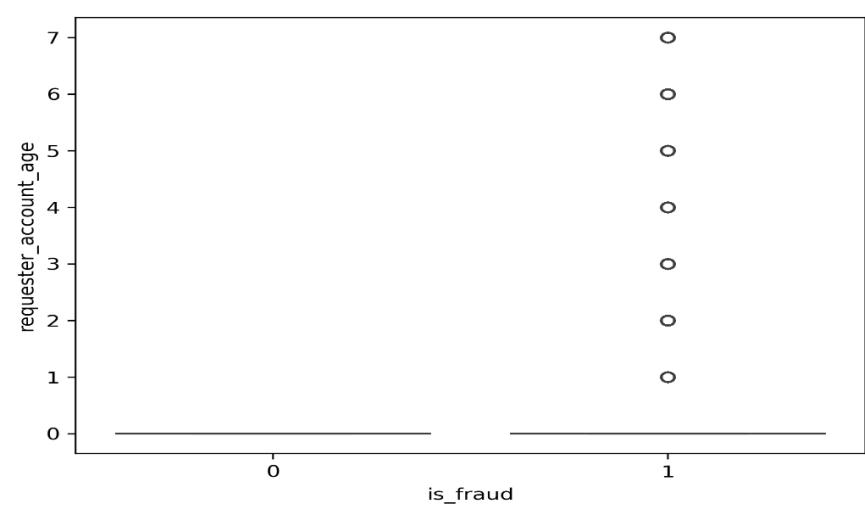


Interpretation

The correlation heatmap illustrates the relationships among numerical variables in the dataset. While some variables exhibit moderate relationships, fraud is influenced by multiple behavioral and transactional characteristics rather than a single highly correlated feature. This supports the use of multi-factor fraud detection models.

Receiver Account Age Analysis

Chart 4:



Interpretation

The analysis shows that newly created receiver accounts have a relatively higher fraud rate compared to older accounts. Limited transaction history and account maturity make these accounts more susceptible to fraudulent activities. Monitoring new receiver accounts can significantly improve fraud detection effectiveness.

Business Recommendations

Based on the statistical analysis, the following recommendations are provided for the fraud prevention team:

1. Implement real-time monitoring for high-risk transactions.
 2. Enable Multi-Factor Authentication (MFA) for suspicious transactions.
 3. Monitor unusual devices, unfamiliar IP addresses, and unexpected locations.
 4. Detect rapid transaction bursts and repeated authentication failures.
 5. Closely monitor newly created receiver accounts until sufficient transaction history is established.
 6. Implement a risk-based fraud scoring model that combines multiple fraud indicators instead of relying solely on transaction amount.
 7. Continuously update fraud detection models using recent transaction data to adapt to evolving fraud patterns.
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Conclusion

This statistical analysis successfully identified several important fraud indicators and suspicious user behaviors within the UPI Fraud Detection dataset. The findings demonstrate that fraud detection should rely on a combination of transaction behavior, authentication patterns, device information, location details, and account characteristics rather than a single feature. Implementing the recommended strategies can improve fraud detection accuracy, reduce false positives, and strengthen the overall security of digital payment systems.